Rai Mai Phao (Zero-Burn Farm) — Team Briefing

*Simple English (A2–B2). All prices and numbers are **our best guess** for the prototype. We still need to test a real machine.*

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1. The project in a few words

Farmers burn crop waste because it is **the cheapest way** to clean the field. We want to make crop waste **worth money**. Our machine **KhumKha** ("worth it" in Thai) sits in a **village station**. It turns crop waste into **animal food, fuel blocks or compost**. Farmers sell with **LINE**. A **driver from the village** picks up the waste from the farm, like Grab. **Farmers get money, drivers get work, buyers get cheap material, and everyone gets cleaner air.**

2. Numbers to remember

The problem (GISTDA, 2026)

What	Number
Farm land burned in Thailand, 2026	**26.0 million rai** (a record)
North	**16.17 million rai** (the most, and the most lung cancer)
Isan (Northeast)	5.59 million rai
Central	1.93 million rai (corn burning up 760%)
West	1.65 million rai (rice burning up 40 times)
East	0.63 million rai (lung cancer going up fastest)
South	almost no burning (the smoke comes from Indonesia)
Less PM2.5	about **8 g for every 1 kg** not burned (our guess)

The machine: KhumKha

What	Number
Name	**KhumKha** (คุ้มค่า) = "worth it"
Size	**85 × 180 × 60 cm**, about a water dispenser
Cutter	about 3 kW
A	animal food or fuel pellets · we get back **85%** · about 60 min for 100 kg
B	fuel blocks (instead of charcoal) · **80%** · about 75 min for 100 kg (+15 min to warm up)
C	compost or mushroom material · about **90%** · about 15 min for 100 kg

What	Number
Check and repair	every **200 hours** of work
Price of one machine	about **200,000 baht** (150,000–280,000)

Prices

What	Price
We buy: corn waste, rice straw, fruit tree branches	**1 baht/kg**
We buy: sugarcane leaves	0.80 baht/kg
15% wet or less	full price
15–20% wet	10% less
More than 20% wet	**we can't take it** (dry it at the station first)
Fee if the farmer takes the product home	A 1.20 · B 1.50 · C 0.40 baht/kg
Pickup fee	**50 baht + 10 baht/km + 0.10 baht/kg**, up to 10 km
When can you ask for a pickup?	**300 kg or more**, or a full truck
Farmer pays for pickup (when selling to us)	**30% of the sale or less**. Our **fund** pays the rest.
Driver gets	**90%** of the full pickup fee, always
We keep	**10%** of the pickup fee

Examples

- **1,000 kg of dry corn waste, 6 km away:** sale 1,000 baht · pickup 210 baht → **farmer gets 790 baht** · driver gets 189 · we keep 21.
- **300 kg, 6 km away:** sale 300 baht · pickup 140 baht · farmer pays 90 · our fund pays 50 → **farmer gets 210 baht** · driver gets 126.

Business

What	Number
Time to pay back the machine (example)	400 kg a day × 25 days = 10 tons a month. About 1.2 baht profit per kg = 12,000 baht a month → **about 17 months**
Village station	**about 2 rai**, like a gas station
Cost of our software now	**0 baht a month**
SDGs	**3, 8, 12, 13** (also 7 and 15)

3. How it works (5 steps)

1. **Ask in LINE:** the farmer says what waste they have and how much. They bring it, or ask for a pickup.
2. **Village pickup:** a driver from the village takes the job and picks it up from the farm.
3. **Check at the station:** we weigh it, take a photo and check if it's dry. We scan the farmer's QR code.
4. **Make the product:** A, B or C.
5. **The farmer chooses:** sell it to us, let us sell it for them, or take it home. They get a receipt in LINE.

The station has 8 parts: scale for trucks · drying area with a roof · machine room · solar roof · storage · shop and office · fire safety · parking. We start with **one machine in each village** and add more later.

4. How the machine works (top to bottom)

Height	Part	What it does
140–180 cm	opening with a safety lid, hand guard, magnet, wet-meter, touch screen, QR scanner, camera	takes in the waste and checks it
100–140 cm	cutter + 2 screens	cuts the waste into small pieces
85–100 cm	moving gate	sends it to A, B or C
30–85 cm	A pellet press · C straight path · B hot press + cutter	makes the product
0–30 cm	3 trays with scales	weighs the product
Back	dust fan and bag, power box, air holes	keeps it safe, clean and cool

Limits: it makes one product at a time (we use a queue) · when it changes from fuel to animal food, it must clean itself first · if the cutter breaks, the machine stops (we keep extra blades, and the system calls a technician).

5. Business model (BMC)

Part	What we have
Customers	farmers who sell waste · people who buy our 6 products · drivers
What we give them	farmers: money and pickup from the farm · buyers: material every month that they can check · village: clean air

Part	What we have
How we reach them	LINE · the drying area in the village · village leaders
Relationship	LINE · clear receipts · a public dashboard
How we make money	**selling products (most money)** · a fee if farmers take products home · 10% of each pickup fee · later: company sponsors, local government, carbon credits
Main work	buy, make, sell · run LINE · take care of the machines
What we need	the machine · LINE system and data · drivers · our brand
Partners	village groups · local government · the university · mushroom farms, cow farms, longan dryers · drivers
Costs	machine (about 200,000 baht) · electricity (less with solar) · buying waste · repairs · workers · pickup fund

Who gets something: farmers (money) · village drivers (extra money) · buyers (cheap material every month) · the village (less smoke, fewer sick people) · local government (a way to stop burning, with real data) · nature (less carbon, waste goes back to the soil). **People, Planet, Profit:** People = money for farmers and drivers · Planet = less smoke and carbon · Profit = selling products and fees.

6. SDGs

SDG	How we help
3 Good health	no burning → less smoke → fewer people with lung problems
8 Good jobs	farmers get money from waste; village people get jobs as drivers
12 No waste	crop waste becomes animal food, fuel and compost
13 Climate	less carbon from burning; we count it on the dashboard
Also 7 · 15	clean fuel + solar roof · fewer forest fires, waste goes back to the soil

7. Scripts

30 seconds (people walking past)

> **"Do you know the smoke in the dry season?"** *(wait for a nod)* This year, Thailand burned **26 million rai** of farm land. That's a record. Farmers burn because it is the cheapest way. So we made **KhumKha**. In Thai, it means **"worth it"**. It turns crop waste into **animal food, fuel or compost**, and you use it with **LINE**. Farmers get money, drivers get jobs, and everyone gets

> cleaner air. We help **SDG 3, 8, 12 and 13**. **"Do you want to try? Scan this QR. It takes 10 seconds."**

2 minutes (people who stop)

1. **Board 1 (20 s):** "The North burned 16 million rai. That's the most in Thailand, and it has the most lung cancer. Farmers know burning is bad. But it's the cheapest way."
2. **Board 2 + iPad (40 s):** "This is KhumKha. It means 'worth it'. You put crop waste in the top. It cuts it, then makes A: animal food, B: fuel blocks, or C: compost. It weighs it, takes a photo, checks if it's dry, and sends a receipt to LINE." *(give them the iPad: X-ray, Show workflow)*
3. **Board 3 (40 s):** "Each village has a station, like a gas station. The farmer asks in LINE. A village driver picks it up, like Grab. We pay about 1 baht a kilo. The farmer pays 30% of the sale or less for the pickup."
4. **End (20 s):** "Farmers get money, drivers get work, buyers get cheap material, everyone gets cleaner air. That's SDG 3, 8, 12 and 13." *(give them the phone)* "Try selling some waste in LINE."

4 minutes (judges)

Part	Show	Say	Time
Start	boards	"We are Rai Mai Phao. We make crop waste worth more than burning it."	10 s
Problem	board 1	"Thailand burned 26 million rai in 2026. That's a record. The North burned 16 million and has the most lung cancer. Farmers burn because it's the cheapest way."	30 s
Machine + demo	board 2 + **give the iPad to the judge**	"This is KhumKha. It means 'worth it'. It cuts crop waste and makes animal food, fuel blocks or compost. It weighs it, checks it, and sends a receipt to LINE." *(Show workflow, X-ray)*	50 s
System	board 3	"Each village has a station, like a gas station. Farmers ask in LINE. A village driver picks it up, like Grab. The farmer pays 30% or less for the pickup. Our fund pays the rest."	30 s
Money	iPad → **Business Model**	"We make money in three ways: we sell the products, we take a small fee, and we keep 10% of each pickup fee. One machine costs about 200,000 baht. We can pay it back in about 17 months."	40 s
SDGs	SDG boxes on board 2	"SDG 3: cleaner air. 8: jobs and money. 12: no waste. 13: less carbon. Good for people, nature and business."	20 s
End	phone with LINE	"Do you want to try? Sell some waste in LINE. You get the receipt in a few seconds."	20 s

Always say: *"These numbers are our best guess. Next, we want to test a real machine."*

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8. Questions and easy answers

The problem and the idea

- **Why do farmers burn?** It's the fastest and cheapest way to clean the field. Ploughing costs money and gives nothing back.
- **Why not stop burning with a law?** There are laws, but farmers still burn if there is no cheaper way. We give them money, so they have a reason to stop.
- **Who are your customers?** Two groups: farmers who sell waste, and people who buy our products (mushroom farms, cow farms, animal farms, factories that need fuel).

The machine

- **What is the machine called?** KhumKha. It means "worth it" in Thai.
- **What does it make?** A: animal food or fuel pellets. B: fuel blocks. C: compost or material for mushrooms.
- **How fast is it?** About 15 minutes for 100 kg of C, 60 minutes for A, 75 minutes for B.
- **What if the waste is wet?** If it's more than 20% wet, we can't take it. The farmer can dry it at the station.
- **What if it breaks?** We keep extra blades. The system calls a technician every 200 hours.
- **Is it safe?** Yes. It has a safety lid, a hand guard, a magnet for metal, a red stop button and a dust fan.
- **Did you build it?** We made a full 3D design and working software. Next, we want to build a real one.

Money

- **How do you make money?** We sell the products. Farmers pay a small fee to take products home. We keep 10% of each pickup fee.
- **How much is the machine?** About 200,000 baht (150,000–280,000). The station cost depends on the village.
- **When do you get the money back?** In about 17 months, if we use 400 kg a day and make about 1.2 baht per kg. This is a guess.
- **Why 1 baht a kilo?** It's better than burning (0 baht), and we still make a profit.
- **Who pays the pickup fund?** It comes from our product sales. It's only for farmers who sell to us.
- **Why 30%?** So the pickup fee doesn't take most of a small farmer's money.
- **Can people ask for very small pickups?** No. You need 300 kg or a full truck. We tell small farms to share a trip.
- **Who are your competitors?** Big power plants buy crop waste, but only near the plant and in big amounts. We go to small farmers in the village.

Impact, technology and law

- **Which SDGs?** 3, 8, 12, 13 (also 7 and 15).
- **How do you measure your impact?** Every job saves the data: kg not burned, less PM2.5 (about 8 g/kg), money for farmers and drivers. You can see it on the dashboard.
- **Is 8 g/kg a real number?** It's our guess. We will check it with research.
- **Why LINE?** Farmers already use LINE every day. They don't need a new app.
- **What about personal data?** Farmers say yes first. We keep only the data we need. We hide names in public. Farmers can delete their data.
- **Can you sell the animal food?** First, we use it on farms in the village. To sell it, we need a permit.
- **Can you protect your brand?** Yes. We can register "Rai Mai Phao" as a trademark.

Hard questions

- **What is your biggest risk?** We haven't tested the numbers yet. Next, we want to build one real machine.
- **What would you do with money?** Build one machine in a village in Chiang Rai, test it, and fix our numbers.
- **What did you learn?** The problem is not that farmers don't know. The problem is money. So we make the good choice the cheaper choice.

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9. Before we open

- [ ] iPad on **/ipad/3d**: model loaded, English, full screen
- [ ] Computer screen on **/present**, press **F**
- [ ] Phone: try one sale in LINE and get the receipt
- [ ] Wi-Fi + phone hotspot
- [ ] Charge the iPad and phone · bring a power strip
- [ ] Practise the 30-second and 4-minute versions with a timer
- [ ] Practise giving the iPad to someone and opening **Business Model**

Links: sustainable-final.vercel.app · /ipad/3d · /ipad/bmc · /present